

GlucoTwin

A digital twin that warns people with Type 2 Diabetes of a blood-sugar spike two hours before it happens

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Team I Dont Think We Can Code | XLRI, PGD-BM 2025-27

Spikes are discovered after the damage is done

101 million

Indians live with diabetes, and 136 million more have prediabetes (ICMR-INDIAB study, 2023)



Spikes above 180 mg/dL after meals

Over years they damage blood vessels, nerves, eyes and kidneys.



Common with carb-heavy Indian meals

Rice-heavy lunches and late dinners push glucose up daily.



Found out too late

A glucometer reading or a lab report weeks later, not a warning beforehand.

What if the warning came two hours before the spike, while there is still time to act?

One virtual copy of each patient that does three things



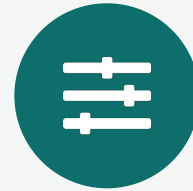
Predict

Every 15 minutes: the chance that glucose crosses 180 mg/dL in the next 2 hours.



Explain

The reasons in plain language, e.g. 95 g of carbs 40 minutes ago, no walk, 5 hours of sleep.



Simulate

What happens if the patient walks for 30 minutes or halves the rice? See the curve before it happens.



Built for diabetologists and primary-care doctors who monitor many patients remotely; a patient app is the next step.

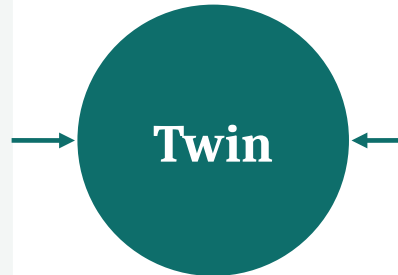
The twin fuses two data streams

All 100 patients are synthetic, as the challenge requires (DPDP Act, HIPAA)

Health record

Static, from the hospital file

- Age, sex, BMI, years with diabetes
- HbA1c, fasting glucose, cholesterol, creatinine
- Medication, blood pressure, past diagnoses
- Family history and TCF7L2 genetic risk marker
- Diet pattern and activity level

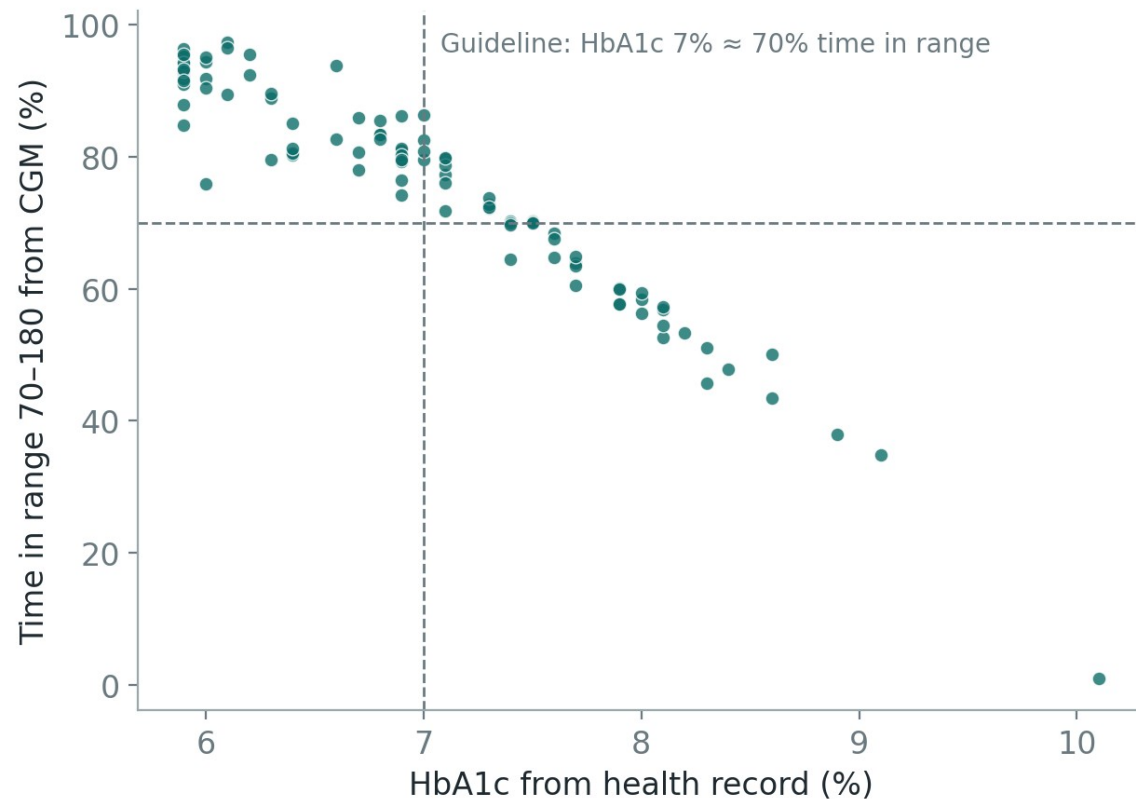


Wearables

Dynamic, every 5 minutes for 14 days

- Continuous glucose monitor
- Heart rate and steps
- Sleep stages: light, deep, REM
- Overnight heart-rate variability
- Meals logged in an app (carbs)

Synthetic, but behaving like real patients



Health records and wearable data agree: patients with worse HbA1c spend less time in the healthy range.

Indian routines

Rice or wheat diets, heavy lunch, evening chai, late dinner, weekend and festival meals.

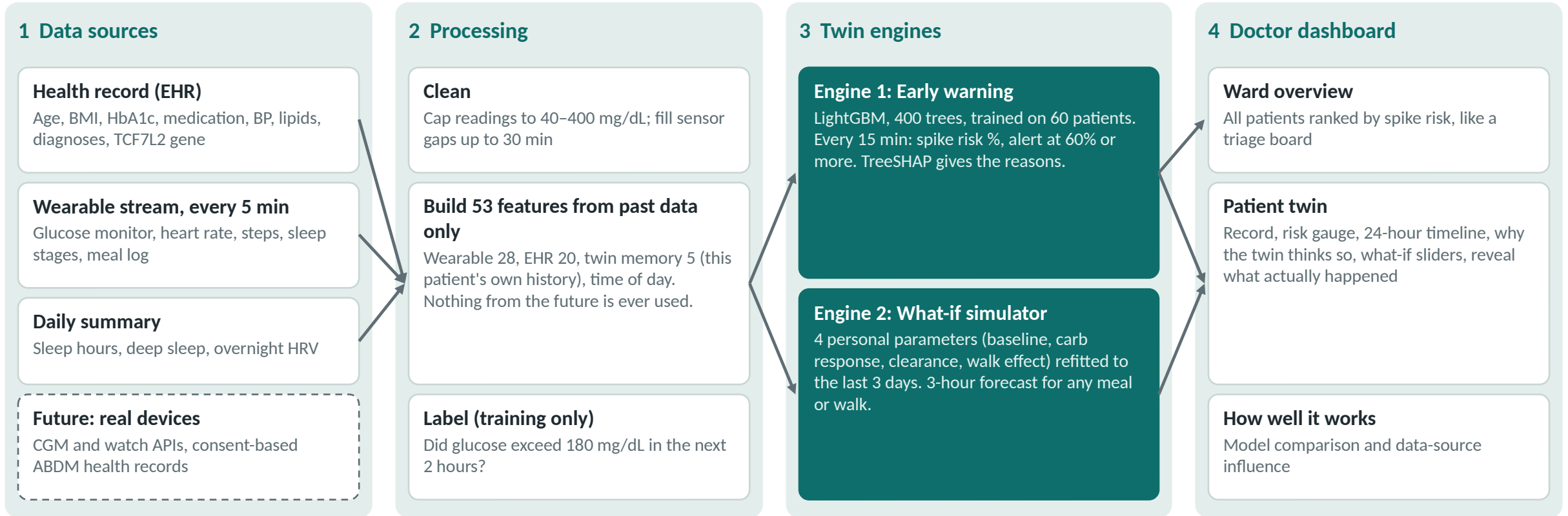
Real-life mess

15% of meals unlogged, portions misjudged by $\pm 20\%$, sensor noise and signal gaps.

Checked against medicine

Average glucose matches the ADAG formula from HbA1c; time in range vs HbA1c correlation -0.96 ; median 5,200 steps and 6.8 h sleep.

How it works: a hybrid twin with two engines



Evaluation: split by patient 60 train / 10 validation / 30 test; AUROC, spikes caught, warning time, false alarms per day. **Deployment:** Colab notebooks (seeded, reproducible) → GitHub → Streamlit Community Cloud. Synthetic data only (DPDP Act, HIPAA).

Engine 1: early warning, tested honestly

28

Wearable clues

Glucose now and its trend, recent carbs, steps, heart rate, last night's sleep and HRV

20

Health record

HbA1c, BMI, medication, blood pressure, lipids, kidney function, genetics

5

Twin memory

How much this patient's glucose usually rises per gram of carbs, learned from their own past

LightGBM

Model

400 decision trees; TreeSHAP explains every single prediction



Tested on 30 patients it never saw

Split by person: 60 train, 10 to set the alert level, 30 test.



No peeking at the future

Only data available at that moment; today's total steps are never used.

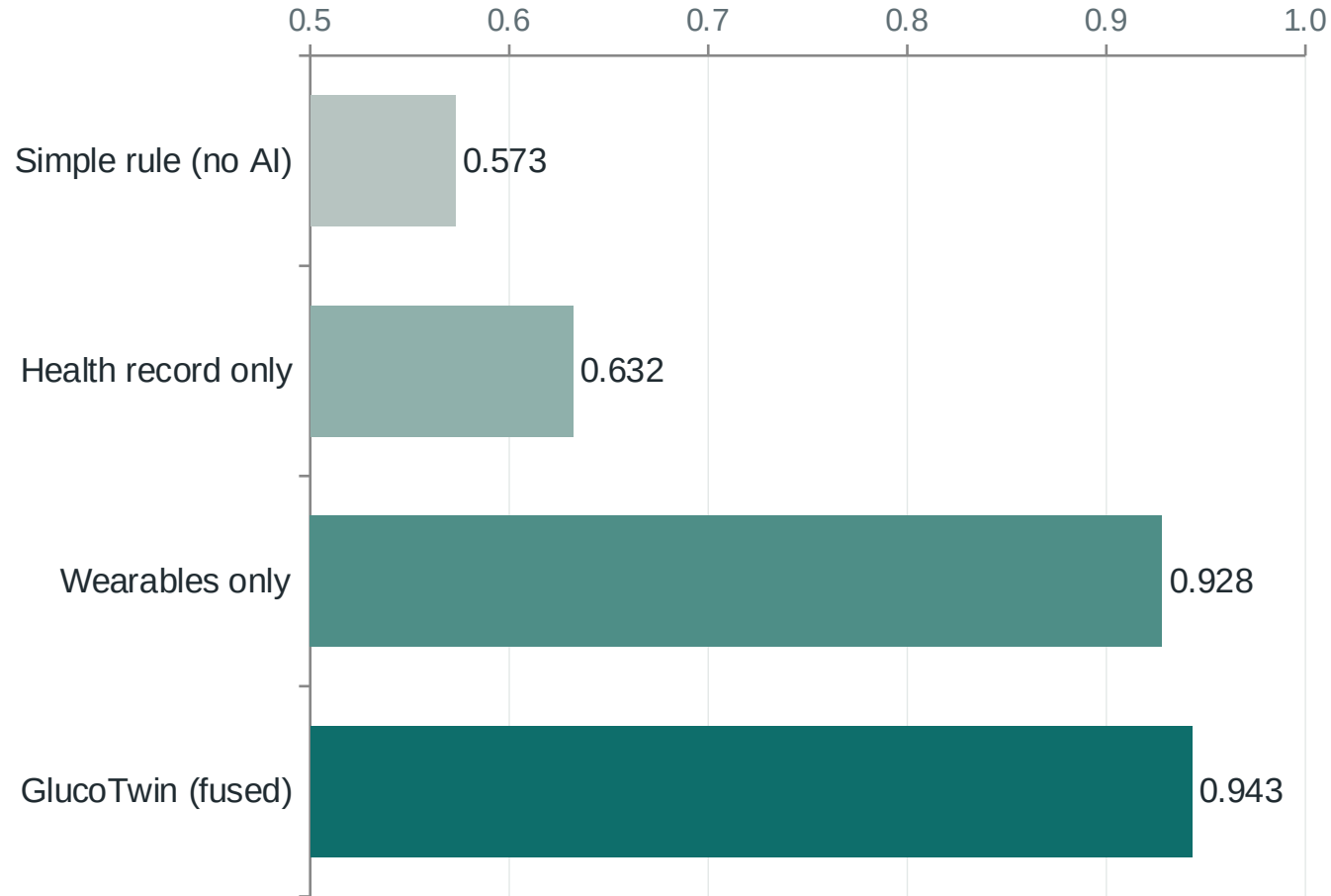


Warns before, not after

No predictions while glucose is already above 180.

Fusing both streams beats either one alone

AUROC on 30 unseen patients (0.5 = coin toss, 1.0 = perfect)



97%

of spikes flagged in advance

~95 min

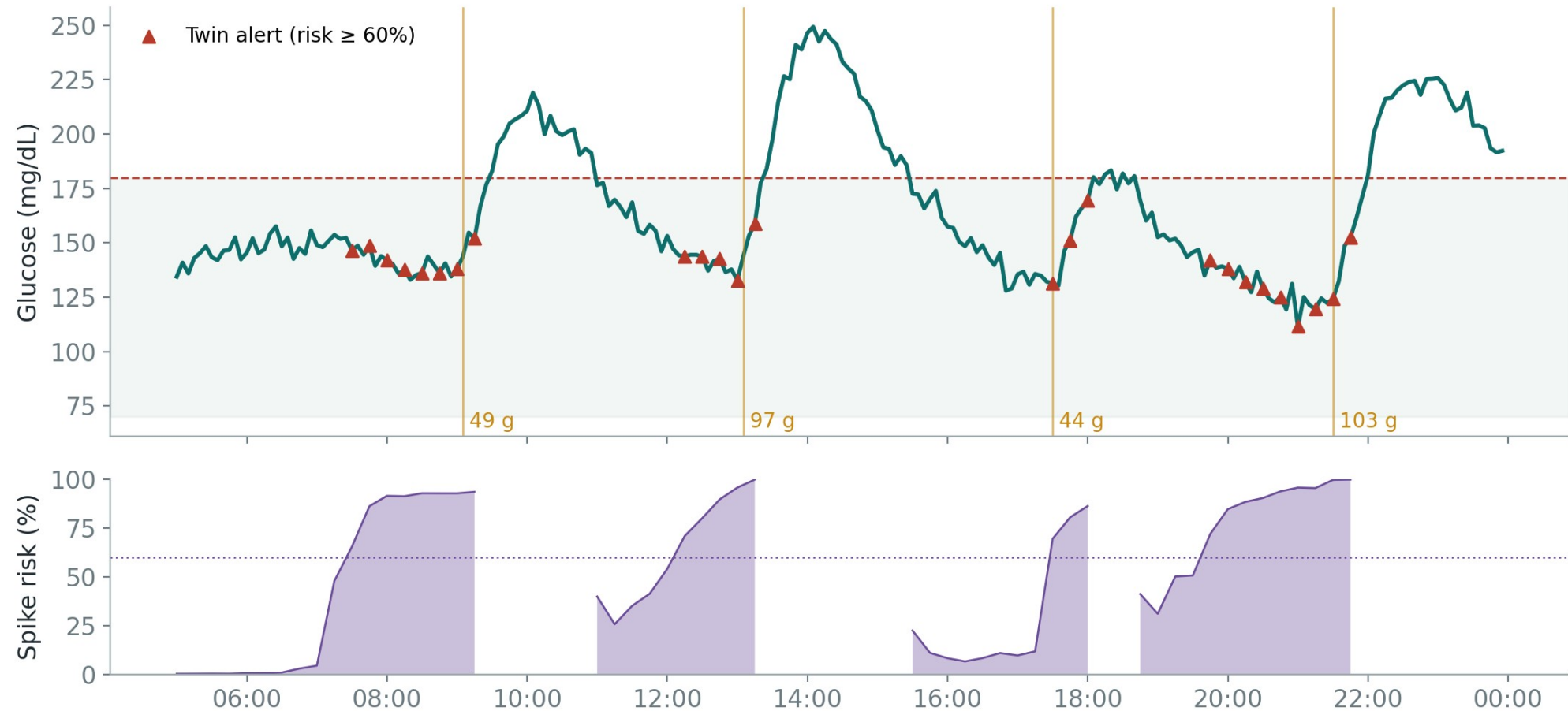
median warning before glucose crosses 180

0.7

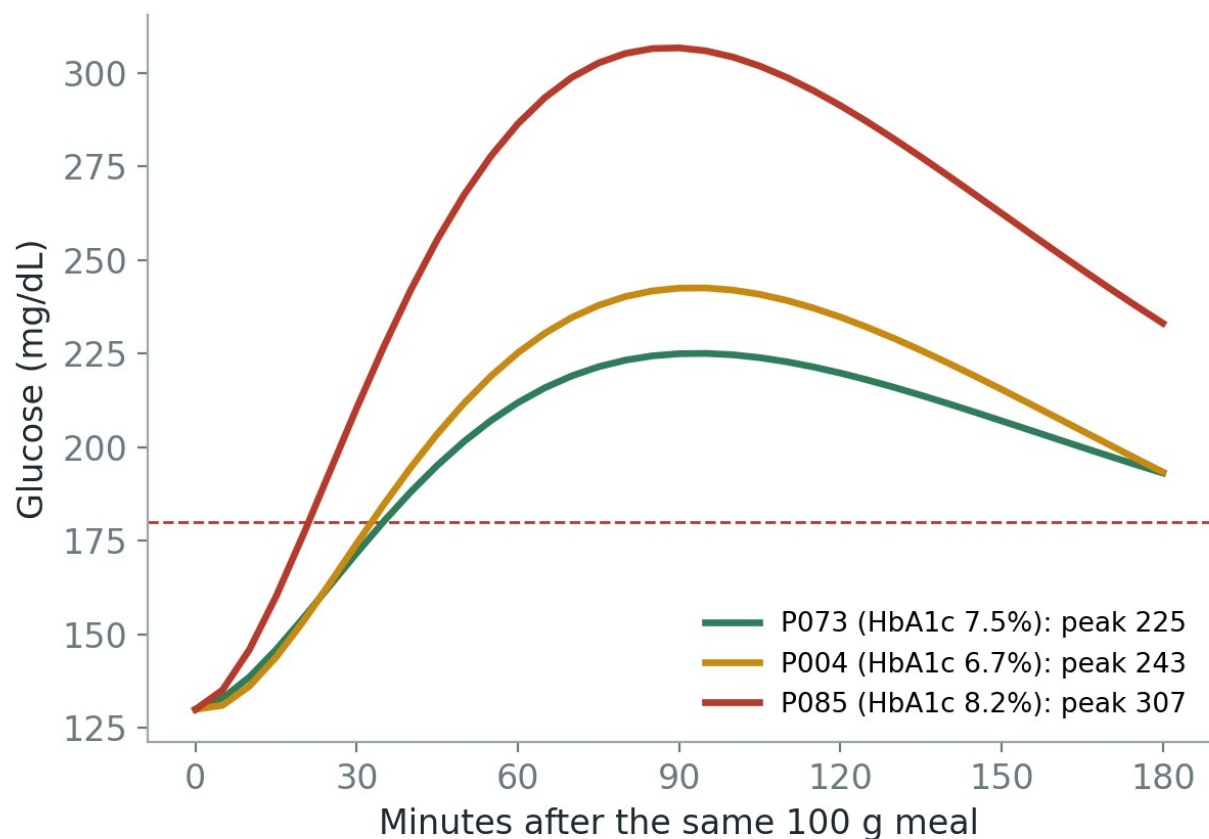
false alarms per patient per day

A day in the life of patient P077

Red triangles: twin alerts. Orange lines: logged meals. The alerts come before each crossing of 180.



Same meal, different bodies: why each patient needs a twin



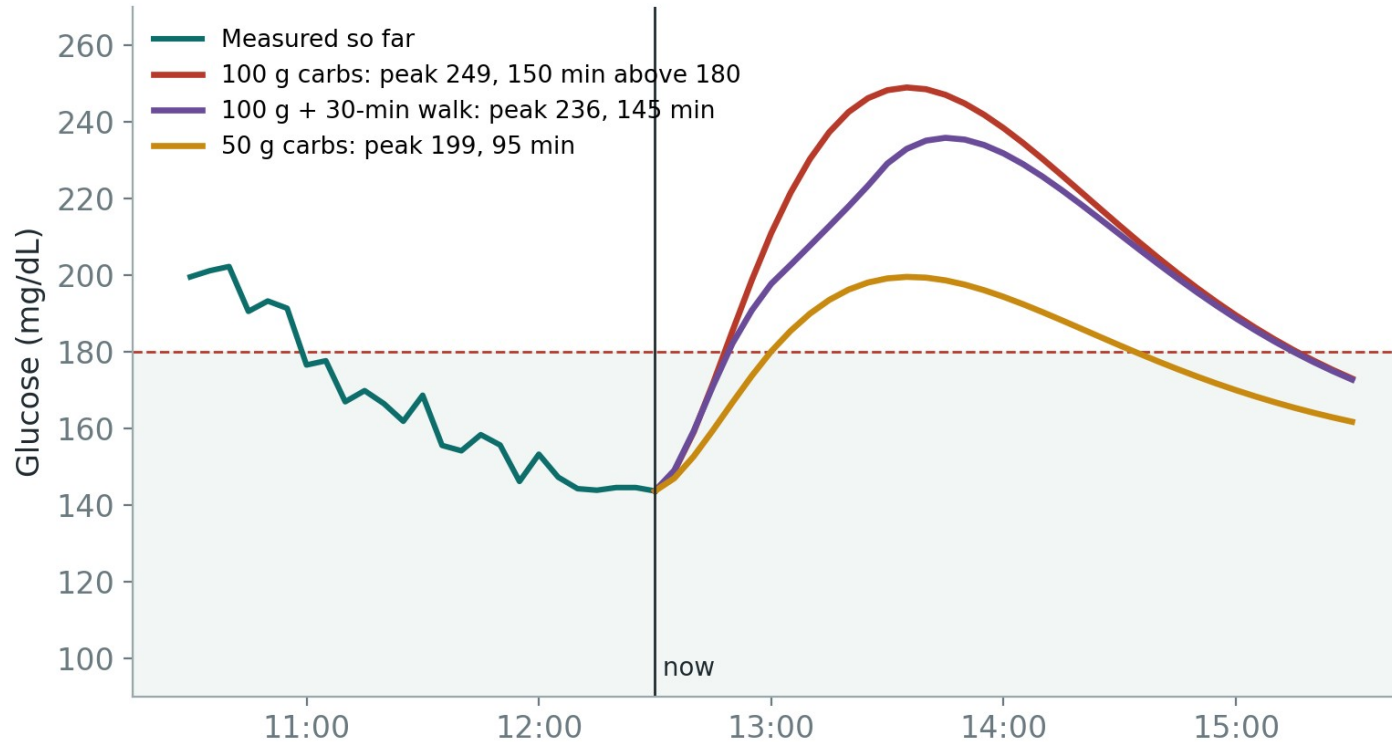
The twin learns four facts about each body from the last 3 days

- Normal glucose level
- Rise per gram of carbs
- How fast sugar is cleared
- How much walking helps

P004 has a better HbA1c than P073, yet spikes higher after the same 100 g meal. A lab report alone would miss this; the twin doesn't.

Engine 2: the what-if simulator

Patient P077 at 12:30, just before lunch



Forecast accuracy on 30 unseen patients

12 mg/dL

typical error 1 hour ahead (vs 23 if you assume no change)

14 mg/dL

typical error 2 hours ahead (vs 37)

Why a physiology model? Machine learning finds patterns; it can't be trusted to answer "what would happen if...". A small model of the body can.

The doctor dashboard is live

idontthinkwecancode.xlri.streamlit.app



Ward overview

All patients ranked by spike risk, like a triage board.



Patient twin

Health record, risk gauge, 24-hour timeline, and why the twin thinks so.



What if...

Meal and walk sliders with a 3-hour forecast; reveal what actually happened.



How well it works

Model comparison and which data sources drive decisions.



Clinic clock

Replays 14 days as if live; the twin only ever sees the past.



Scan to try it yourself. No installation,
works on a phone.

Built to be trusted by doctors



Unseen patients

All results come from 30 people the model never trained on.



No future data

Every prediction uses only what was known at that moment.



Alarm fatigue measured

Under 1 false alarm per patient per day at the chosen alert level.



Explainable

Every risk score shows its top reasons, labelled by data source.



Actionable

What-if forecasts turn a warning into a concrete choice: walk, eat less, or both.



Privacy by design

Synthetic data only; ready for consent-based ABDM records later.

Honest limits, and the road to real patients

What we know it can't do yet

- Synthetic patients follow steadier routines than real people, so the twin partly anticipates meals from the time of day.
- Real-world accuracy will be lower than on synthetic data.
- What-if forecasts are decision aids, not treatment instructions.

Next steps

- Validate on public CGM datasets (OhioT1DM, Shanghai T2DM).
- Stress-test with irregular routines and unlogged meals.
- Pilot with a diabetes clinic, doctor in the loop.
- Connect to affordable CGMs and ABDM health records.
- Patient app with nudges in Indian languages.

Thank you

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Live dashboard: idontthinkwecancodexlri.streamlit.app
Code: github.com/b25474-sudo/IDontThinkWeCanCode_XLRI



Dashboard



GitHub